Supplementary Material

Log-Transformed Red-Tailed Hawk Optimization for Single-Diode Photovoltaic Parameter Extraction

Salma El Aamery and Rachid El Bachtiri
Laboratory of Innovative Technologies and Computer Science,
Higher School of Technology, Sidi Mohamed Ben Abdellah University,
Fez, Morocco

This document reports the complete equal function-evaluation (FE) results used in the revised manuscript. Each algorithm receives exactly 149050 objective evaluations per run, with $N = 50$ and 30 matched runs. For dataset d and run r, all algorithms use `rng(50000 + 1000*d + r)`, so the initial population is identical within a matched run. RTH uses 1000 iterations; the single-evaluation algorithms use 2980 iterations; HHO uses an FE-based progress and stopping rule because its evaluation count per iteration is branch-dependent.

The near-zero RTH dispersion is reported as numerical convergence under this protocol, not as proof of a certified global optimum. The MATLAB files `Benchmark_EqualFE.m`, `make_counted_objective.m`, `HHO_FE.m`, and the saved MAT/CSV outputs accompany this document.

Table S1 – Equal-FE statistical summary

Table 1: RMSE summary over 30 matched runs. All RMSE values are in amperes.

Dataset	Algorithm	Best	Mean	Median	Worst	Std. dev.	IQR
SM55	RTH	2.4895e-05	2.4895e-05	2.4895e-05	2.4895e-05	1.3223e-15	4.1578e-16
SM55	PSO	9.4556e-05	5.3792e-03	3.4667e-03	3.8966e-02	7.5056e-03	3.7973e-03
SM55	GWO	2.2286e-03	1.3002e-02	9.9279e-03	3.8047e-02	8.5249e-03	1.2082e-02
SM55	WOA	6.2696e-03	3.0023e-02	3.2372e-02	8.9588e-02	1.8503e-02	2.4725e-02
SM55	HHO	3.8559e-04	8.9227e-03	7.8577e-03	2.5300e-02	6.0071e-03	6.8949e-03
SM55	AVOA	7.9232e-04	3.2379e-02	3.9147e-02	4.8227e-02	1.4875e-02	2.4174e-02
SM55	MGO	1.4508e-03	1.1051e-02	7.4291e-03	4.1254e-02	1.0809e-02	5.5295e-03
RTC	RTH	7.7301e-04	7.7301e-04	7.7301e-04	7.7301e-04	5.8409e-17	8.1098e-17
RTC	PSO	7.7402e-04	1.6457e-03	8.5524e-04	8.1922e-03	2.2324e-03	2.1879e-04
RTC	GWO	1.1246e-03	2.5656e-03	2.2494e-03	5.1445e-03	1.1164e-03	1.2705e-03
RTC	WOA	1.2364e-03	9.4104e-03	8.9918e-03	5.0715e-02	8.9215e-03	6.5886e-03
RTC	HHO	7.7548e-04	2.7494e-03	1.9471e-03	8.1925e-03	2.4639e-03	1.7178e-03
RTC	AVOA	1.3798e-03	6.5526e-03	7.9658e-03	9.4032e-03	2.3048e-03	3.1702e-03
RTC	MGO	1.4956e-03	4.4948e-03	2.8888e-03	8.9927e-03	2.5455e-03	4.7053e-03
PWP201	RTH	2.0530e-03	2.0530e-03	2.0530e-03	2.0530e-03	1.3605e-16	1.0278e-16
PWP201	PSO	2.0535e-03	2.4284e-03	2.2043e-03	7.4937e-03	9.7555e-04	2.8544e-04
PWP201	GWO	2.2897e-03	3.9553e-03	2.8161e-03	1.4012e-02	2.5516e-03	1.4136e-03
PWP201	WOA	2.4653e-03	7.1952e-03	6.5519e-03	2.1394e-02	4.1571e-03	5.5450e-03
PWP201	HHO	2.0539e-03	2.8827e-03	2.6854e-03	7.5891e-03	1.0875e-03	4.6299e-04
PWP201	AVOA	4.4990e-03	8.5450e-03	8.2339e-03	1.1797e-02	1.5430e-03	1.5220e-03
PWP201	MGO	2.4096e-03	4.2282e-03	3.6175e-03	9.2317e-03	1.9308e-03	1.6162e-03

Table S2 – Complete RMSE values: Shell SM55

Table 2: Complete equal-FE RMSE values (A) for Shell SM55 over 30 matched runs.

Run	RTH	PSO	GWO	WOA	HHO	AVOA	MGO
1	2.48954214e-05	4.50430993e-03	3.80472307e-02	3.56345684e-02	2.53000085e-02	2.47754229e-02	5.56000895e-03
2	2.48954214e-05	3.36568353e-03	2.52867978e-02	2.02521018e-02	1.54110118e-02	4.09440392e-02	3.85515030e-02
3	2.48954214e-05	1.56405389e-02	9.78146807e-03	5.17352002e-02	2.10744125e-02	4.62843313e-02	3.97800335e-03
4	2.48954214e-05	2.28772346e-03	1.17718725e-02	3.59267497e-02	1.03397088e-02	4.21083277e-02	1.45082503e-03
5	2.48954214e-05	7.17892049e-03	1.67328569e-02	4.68470214e-02	8.80268689e-03	1.35892085e-02	2.99219794e-03
6	2.48954214e-05	1.30547038e-03	8.70601946e-03	3.64961983e-02	1.05364347e-02	3.69892220e-02	5.61531403e-03
7	2.48954214e-05	4.95835768e-03	1.53435599e-02	2.26418202e-02	1.03436848e-02	4.56847620e-02	4.80918803e-03
8	2.48954214e-05	3.01301137e-03	6.09986697e-03	6.26961750e-03	3.48875394e-03	4.75868406e-02	1.12454788e-02
9	2.48954214e-05	1.31867891e-03	4.27144785e-03	1.41165305e-02	3.24348029e-03	4.38364262e-02	7.31108733e-03
10	2.48954214e-05	1.26393700e-03	2.04082756e-02	3.41372336e-02	9.52169384e-03	4.12511440e-02	6.70612807e-03
11	2.48954214e-05	9.67255853e-05	9.19960907e-03	6.60549852e-03	1.24367088e-02	4.43693985e-02	5.05949979e-03
12	2.48954214e-05	5.48801813e-03	1.13751478e-02	1.03158066e-02	6.83561642e-03	3.03537442e-02	2.15766005e-03
13	2.48954214e-05	4.63378365e-03	2.24491206e-02	2.72882935e-02	9.17449137e-03	1.98956634e-02	8.80990205e-03
14	2.48954214e-05	3.66266744e-03	2.44312396e-03	6.49529083e-03	1.10247121e-03	3.33279048e-02	5.12239267e-03
15	2.48954214e-05	5.91765163e-03	3.93334551e-03	3.34210589e-02	5.14344096e-03	4.07321642e-02	1.37547860e-02
16	2.48954214e-05	3.76056356e-03	2.00106796e-02	7.20382921e-03	9.44122754e-04	4.55199277e-02	9.08532603e-03
17	2.48954214e-05	2.61005015e-03	7.00862204e-03	1.54478836e-02	3.85594745e-04	7.47474101e-03	5.69139641e-03
18	2.48954214e-05	1.91809009e-03	2.22857855e-03	4.00671282e-02	3.88127028e-03	4.47992705e-02	7.39976139e-03
19	2.48954214e-05	4.66864232e-03	1.54098529e-02	3.64429343e-02	7.33001228e-03	4.82269437e-02	9.30876411e-03
20	2.48954214e-05	2.15779592e-03	5.77987010e-03	3.24938188e-02	7.92152204e-03	7.92317244e-04	3.92455599e-02
21	2.48954214e-05	1.39724086e-03	9.77608304e-03	1.86110634e-02	1.45898912e-02	3.24179940e-03	1.76775288e-03
22	2.48954214e-05	9.45559829e-05	2.54291630e-02	5.83288443e-02	2.61330707e-03	1.52229096e-02	8.55045376e-03
23	2.48954214e-05	3.56766550e-03	5.74358685e-03	1.22244905e-02	3.73812752e-03	8.21762308e-03	1.15569445e-02
24	2.48954214e-05	2.20564009e-03	1.00743818e-02	4.54273709e-02	5.89005577e-03	2.08580711e-02	2.35183534e-02
25	2.48954214e-05	1.81211130e-02	2.51919759e-02	4.38753345e-02	6.68345151e-03	3.91785704e-02	8.13466253e-03
26	2.48954214e-05	3.89656120e-02	8.97059843e-03	2.88422825e-02	1.51645023e-02	4.45126235e-02	6.85962940e-03
27	2.48954214e-05	1.04710765e-03	8.45808274e-03	8.95881718e-02	1.12768121e-02	1.97685258e-02	7.45853008e-03
28	2.48954214e-05	7.70109578e-03	2.25391031e-02	3.22500259e-02	1.90086238e-02	3.91160143e-02	4.12544330e-02
29	2.48954214e-05	1.43842179e-03	1.01797477e-02	1.12859614e-02	7.79389502e-03	3.85924669e-02	8.52311730e-03
30	2.48954214e-05	7.08550263e-03	7.41106058e-03	4.04087500e-02	7.70485307e-03	4.41340159e-02	2.00611452e-02

Table S3 – Complete RMSE values: RTC France

Table 3: Complete equal-FE RMSE values (A) for RTC France over 30 matched runs.

Run	RTH	PSO	GWO	WOA	HHO	AVOA	MGO
1	7.73006269e-04	7.76908156e-04	1.47844705e-03	2.45525478e-03	8.19209070e-03	5.41887378e-03	6.23100109e-03
2	7.73006269e-04	2.03559741e-03	2.45324577e-03	6.47585880e-03	2.05134883e-03	7.90346175e-03	8.79287849e-03
3	7.73006269e-04	8.31323320e-04	4.79906201e-03	9.53047490e-03	1.60810878e-03	9.40318124e-03	2.23776225e-03
4	7.73006269e-04	9.17287312e-04	1.12455824e-03	1.25188206e-02	2.46345963e-03	4.89244380e-03	7.98004003e-03
5	7.73006269e-04	9.23206159e-04	1.74541606e-03	1.02870533e-02	1.60997167e-03	8.17193644e-03	7.40030593e-03
6	7.73006269e-04	1.20811033e-03	2.25228167e-03	3.35499278e-03	2.05841062e-03	8.06399859e-03	7.87311197e-03
7	7.73006269e-04	8.19223175e-03	2.64076409e-03	1.57899634e-02	3.55322661e-03	8.25831588e-03	8.99274022e-03
8	7.73006269e-04	8.00550599e-04	4.67987329e-03	1.01678381e-02	2.86036534e-03	7.86840988e-03	2.63277315e-03
9	7.73006269e-04	1.01941071e-03	2.15799282e-03	8.01101237e-03	2.65805751e-03	8.19871686e-03	5.70256353e-03
10	7.73006269e-04	8.19209728e-04	5.07837752e-03	1.10433819e-02	8.73350052e-04	3.67189281e-03	8.86089939e-03
11	7.73006269e-04	7.88740093e-04	1.44954585e-03	3.74332904e-03	3.06920192e-03	8.21182729e-03	2.28142328e-03
12	7.73006269e-04	7.77629225e-04	2.92066289e-03	1.07752706e-02	8.97029025e-04	8.21263038e-03	2.40084996e-03
13	7.73006269e-04	7.80756524e-04	1.95087312e-03	5.07154324e-02	8.39992966e-04	8.22652243e-03	2.26567399e-03
14	7.73006269e-04	7.76815000e-04	2.13727120e-03	9.57034784e-03	1.96505776e-03	5.70762717e-03	2.37016614e-03
15	7.73006269e-04	8.10363338e-04	2.39267463e-03	1.23640478e-03	1.05993331e-03	8.12717117e-03	1.49560554e-03
16	7.73006269e-04	7.79581318e-04	3.26360656e-03	2.47500649e-03	8.69422732e-04	3.01046707e-03	3.12197855e-03
17	7.73006269e-04	8.61219750e-04	1.96690582e-03	9.23811213e-03	6.72266109e-03	2.30276169e-03	2.80838844e-03
18	7.73006269e-04	7.93198273e-04	3.21769056e-03	1.33566746e-02	1.23183533e-03	3.92636767e-03	2.73138502e-03
19	7.73006269e-04	8.19223175e-03	3.35670109e-03	9.57869403e-03	7.82221525e-04	8.18095723e-03	2.84384713e-03
20	7.73006269e-04	8.63889542e-04	1.30547183e-03	9.30638044e-03	8.19253920e-03	8.02823310e-03	2.40800461e-03
21	7.73006269e-04	7.74017315e-04	2.18851000e-03	8.20592823e-03	8.19253342e-03	1.37979971e-03	4.64858369e-03
22	7.73006269e-04	8.19223175e-03	3.30189086e-03	8.74547857e-03	1.91261560e-03	8.27976765e-03	2.34491511e-03
23	7.73006269e-04	9.51826839e-04	2.32604505e-03	4.31576891e-03	1.18811813e-03	8.40401536e-03	2.93367834e-03
24	7.73006269e-04	1.07224968e-03	2.14769939e-03	4.60560753e-03	1.58364221e-03	4.14567916e-03	2.59341805e-03
25	7.73006269e-04	1.16972048e-03	1.20837788e-03	1.07136192e-02	8.19223175e-03	8.11861529e-03	4.97022113e-03
26	7.73006269e-04	7.82048676e-04	2.43525912e-03	3.91927064e-03	7.75475833e-04	2.10870982e-03	2.24302075e-03
27	7.73006269e-04	7.74328427e-04	1.75088376e-03	1.96113215e-02	2.14424495e-03	8.06923332e-03	7.42068445e-03
28	7.73006269e-04	8.49259622e-04	2.24644508e-03	7.58989421e-03	2.21900861e-03	7.80577462e-03	2.58321573e-03
29	7.73006269e-04	9.18422553e-04	5.14454783e-03	1.61828030e-03	1.92922385e-03	5.77577969e-03	5.56374700e-03
30	7.73006269e-04	9.39701011e-04	1.84701551e-03	3.35614752e-03	7.86110057e-04	6.70487679e-03	8.11108570e-03

Table S4 – Complete RMSE values: Photowatt-PWP201

Table 4: Complete equal-FE RMSE values (A) for Photowatt-PWP201 over 30 matched runs.

Run	RTH	PSO	GWO	WOA	HHO	AVOA	MGO
1	2.05296064e-03	2.09341231e-03	4.53075306e-03	3.50501778e-03	2.83347508e-03	7.55496512e-03	3.63221112e-03
2	2.05296064e-03	2.18672266e-03	2.64919224e-03	6.97422558e-03	2.06603889e-03	7.61841048e-03	2.95082317e-03
3	2.05296064e-03	2.07215164e-03	2.39397303e-03	1.19411099e-02	2.68716364e-03	7.66844889e-03	2.59254998e-03
4	2.05296064e-03	2.21056955e-03	3.65446682e-03	7.18861568e-03	2.05390804e-03	7.93419953e-03	7.90665900e-03
5	2.05296064e-03	2.69051168e-03	8.45317601e-03	6.30498027e-03	2.54276396e-03	9.61073610e-03	2.46126520e-03
6	2.05296064e-03	2.25657747e-03	3.08002726e-03	9.31393473e-03	2.40770080e-03	8.55751340e-03	2.40962929e-03
7	2.05296064e-03	2.19805191e-03	2.81275645e-03	9.96862021e-03	2.18002807e-03	7.86615363e-03	2.79476070e-03
8	2.05296064e-03	2.50089391e-03	8.49997498e-03	9.51430375e-03	2.79645548e-03	1.17971594e-02	7.70602894e-03
9	2.05296064e-03	2.05348132e-03	2.28968603e-03	6.14960563e-03	2.52386102e-03	4.49900548e-03	4.16569686e-03
10	2.05296064e-03	2.19568758e-03	3.84708498e-03	4.22627648e-03	4.74278158e-03	9.01363752e-03	3.70874590e-03
11	2.05296064e-03	2.30906664e-03	2.41199903e-03	5.13221733e-03	2.21480384e-03	8.22030240e-03	5.29106277e-03
12	2.05296064e-03	2.06544791e-03	2.78117079e-03	4.03509019e-03	2.15499204e-03	8.44385156e-03	3.44333114e-03
13	2.05296064e-03	7.49371836e-03	2.49126703e-03	8.25072813e-03	2.90476697e-03	7.27031206e-03	2.44194116e-03
14	2.05296064e-03	2.05864844e-03	7.02343105e-03	1.04191091e-02	2.68355582e-03	1.11602559e-02	3.72551856e-03
15	2.05296064e-03	2.35876611e-03	2.67765757e-03	2.80792339e-03	2.69388675e-03	8.27081573e-03	3.60280876e-03
16	2.05296064e-03	2.68876266e-03	3.20325077e-03	2.46530974e-03	2.86341536e-03	8.25229763e-03	3.74401999e-03
17	2.05296064e-03	2.26895640e-03	3.56956521e-03	9.65975591e-03	2.87988574e-03	1.06207303e-02	2.57492160e-03
18	2.05296064e-03	2.05540060e-03	3.96469937e-03	7.79884939e-03	2.93885645e-03	8.19372399e-03	3.30105495e-03
19	2.05296064e-03	2.08828961e-03	3.55706363e-03	1.03392662e-02	7.58911217e-03	9.26208556e-03	4.57594288e-03
20	2.05296064e-03	2.67999337e-03	2.55941302e-03	1.06483867e-02	2.99758365e-03	7.87619321e-03	6.28353212e-03
21	2.05296064e-03	2.21854702e-03	2.50906640e-03	6.50757592e-03	4.81438624e-03	1.17194045e-02	3.64400272e-03
22	2.05296064e-03	2.18951797e-03	5.76994703e-03	4.27082956e-03	2.22222310e-03	6.93248127e-03	9.23166325e-03
23	2.05296064e-03	2.38041949e-03	2.78771236e-03	4.20848319e-03	2.42802251e-03	7.22873188e-03	2.64865123e-03
24	2.05296064e-03	2.06734990e-03	2.33893527e-03	1.42368257e-02	2.91894597e-03	7.59682504e-03	4.36216212e-03
25	2.05296064e-03	2.24944702e-03	4.52684532e-03	6.59624358e-03	2.61174131e-03	7.88791556e-03	2.89149560e-03
26	2.05296064e-03	2.16292367e-03	1.40121285e-02	3.89058831e-03	2.70842683e-03	7.70642689e-03	7.10402480e-03
27	2.05296064e-03	2.40631263e-03	2.81936779e-03	2.13941150e-02	2.11960591e-03	1.04835484e-02	3.21171901e-03
28	2.05296064e-03	2.15211784e-03	2.73106435e-03	2.66141558e-03	2.61351913e-03	8.61656134e-03	8.00764595e-03
29	2.05296064e-03	2.43760248e-03	2.38056743e-03	2.97082795e-03	2.68777637e-03	8.24744863e-03	3.21207044e-03
30	2.05296064e-03	2.06318658e-03	2.33266103e-03	2.47604954e-03	2.60239765e-03	1.02405495e-02	3.21900804e-03

Table S5 – Paired statistical comparisons

Table 5: Paired Wilcoxon signed-rank comparisons of RTH with each competitor under matched initial populations.

Dataset	Comparison	p	Holm p	Median Δ	RTH wins	Competitor wins
SM55	RTH vs. PSO	1.734e-06	1.041e-05	3.4418e-03	1.00	0.00
SM55	RTH vs. GWO	1.734e-06	1.041e-05	9.9030e-03	1.00	0.00
SM55	RTH vs. WOA	1.734e-06	1.041e-05	3.2347e-02	1.00	0.00
SM55	RTH vs. HHO	1.734e-06	1.041e-05	7.8328e-03	1.00	0.00
SM55	RTH vs. AVOA	1.734e-06	1.041e-05	3.9122e-02	1.00	0.00
SM55	RTH vs. MGO	1.734e-06	1.041e-05	7.4043e-03	1.00	0.00
RTC	RTH vs. PSO	1.734e-06	1.041e-05	8.2233e-05	1.00	0.00
RTC	RTH vs. GWO	1.734e-06	1.041e-05	1.4764e-03	1.00	0.00
RTC	RTH vs. WOA	1.734e-06	1.041e-05	8.2188e-03	1.00	0.00
RTC	RTH vs. HHO	1.734e-06	1.041e-05	1.1741e-03	1.00	0.00
RTC	RTH vs. AVOA	1.734e-06	1.041e-05	7.1928e-03	1.00	0.00
RTC	RTH vs. MGO	1.734e-06	1.041e-05	2.1158e-03	1.00	0.00
PWP201	RTH vs. PSO	1.734e-06	1.041e-05	1.5135e-04	1.00	0.00
PWP201	RTH vs. GWO	1.734e-06	1.041e-05	7.6310e-04	1.00	0.00
PWP201	RTH vs. WOA	1.734e-06	1.041e-05	4.4989e-03	1.00	0.00
PWP201	RTH vs. HHO	1.734e-06	1.041e-05	6.3240e-04	1.00	0.00
PWP201	RTH vs. AVOA	1.734e-06	1.041e-05	6.1809e-03	1.00	0.00
PWP201	RTH vs. MGO	1.734e-06	1.041e-05	1.5645e-03	1.00	0.00

Δ is competitor RMSE minus RTH RMSE. The signed-rank values are supporting evidence because the RTH samples have negligible dispersion at the displayed precision.